# VANISHING OF MOD P ETALE COHOMOLOGY OF SCHEMES OF CHARACTERISTIC P ABOVE A DIMENSION BOUND

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument for both parts of the statement.

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*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- Artin-Schreier sequence in characteristic p. Same chapter, the Artin-Schreier short exact sequence giving the cohomology with coefficients Z/p in terms of the structure sheaf and the Frobenius. Supplies the computation reducing mod p cohomology to coherent cohomology.
- Vanishing of the cohomology of quasi-coherent sheaves on affine schemes and the dimension bound for quasi-compact quasi-separated schemes. Same chapter together with the theory of coherent cohomology, the vanishing of the cohomology of quasi-coherent sheaves on affine schemes in positive degrees and the general dimension bound. Supplies the two assertions of the statement.

*Source reference.* Original theorem 03P2: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2280–2280. Navigation pointer invoked at proof line 9627; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 01XB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9631–9631. Navigation pointer invoked at proof line 9631; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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*Date*: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original proposition 0A3G: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9634–9634. Navigation pointer invoked at proof line 9634; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 094L: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9637–9637. Navigation pointer invoked at proof line 9637; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $p$ be a prime. Let $S$ be a scheme of characteristic $p$.*

(1) *If $S$ is affine, then $H^q_{étale}(S, \underline{\mathbf{Z}/p\mathbf{Z}}) = 0$ for all $q \geq 2$.*
(2) *If $S$ is a quasi-compact and quasi-separated scheme of dimension $d$, then $H^q_{étale}(S, \underline{\mathbf{Z}/p\mathbf{Z}}) = 0$ for all $q \geq 2 + d$.*

*Proof.* Recall that the étale cohomology of the structure sheaf is equal to its cohomology on the underlying topological space (Theorem 03P2). The first statement follows from the Artin-Schreier exact sequence and the vanishing of cohomology of the structure sheaf on an affine scheme (Cohomology of Schemes, Lemma 01XB). The second statement follows by the same argument from the vanishing of Cohomology, Proposition 0A3G and the fact that $S$ is a spectral space (Properties, Lemma 094L). □